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SET 6 – PRODUCT SALES ANALYSIS

Aim

To analyze monthly product sales for the first quarter using a grouped bar chart, stacked area chart, and table in R.

Algorithm

1. Create the product sales dataset for January, February, and March.
2. Convert the data into a suitable format for visualization.
3. Create a grouped bar chart to compare sales of each product.
4. Create a stacked area chart to show the overall sales trend.
5. Display the monthly sales data as a table.
6. Add suitable labels and titles to all charts.

R Program

```
library(ggplot2)
```

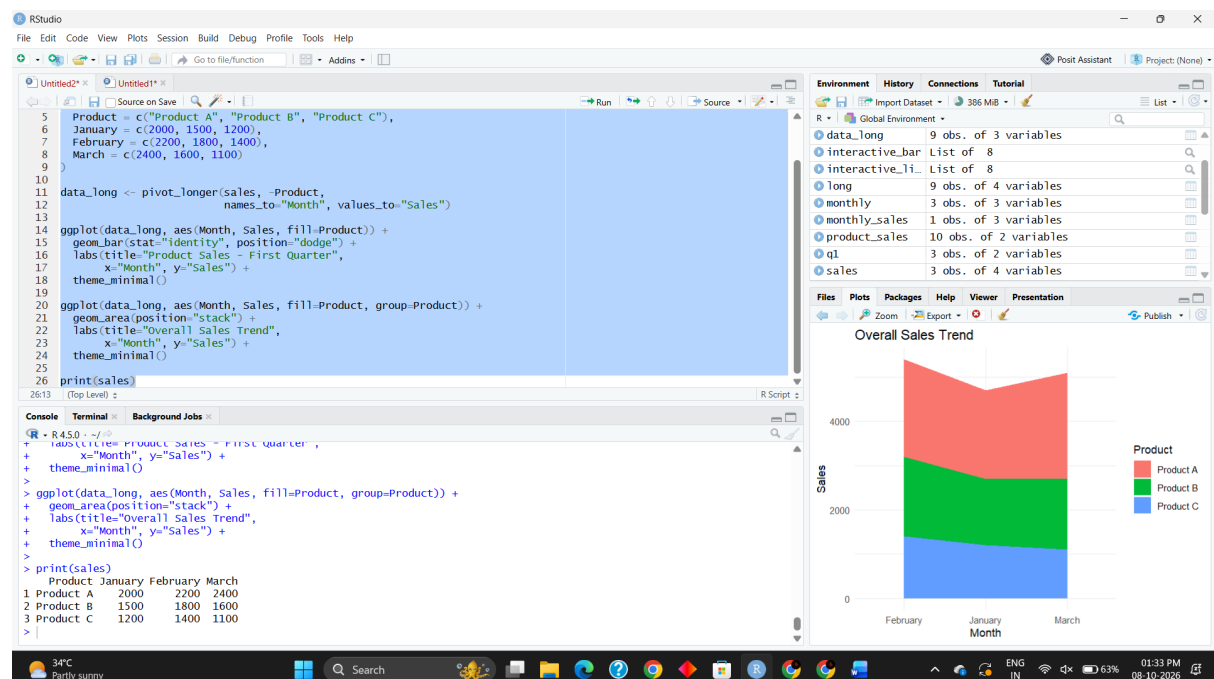
```
library(tidyr)
```

```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)  
  
data_long <- pivot_longer(sales, -Product,  
  names_to="Month", values_to="Sales")  
  
ggplot(data_long, aes(Month, Sales, fill=Product)) +  
  geom_bar(stat="identity", position="dodge") +  
  labs(title="Product Sales - First Quarter",  
    x="Month", y="Sales") +  
  theme_minimal()
```

```
ggplot(data_long, aes(Month, Sales, fill=Product, group=Product)) +
  geom_area(position="stack") +
  labs(title="Overall Sales Trend",
        x="Month", y="Sales") +
  theme_minimal()

print(sales)
```

output:



Result

The grouped bar chart successfully compares the sales of Products A, B, and C for January, February, and March. The stacked area chart shows the overall sales trend, and the table displays the monthly sales data for each product.